
Introduction to Artificial Intelligence

Probabilistic Reasoning

Probabilistic Reasoning

- Logic is used to represent and reason knowledge about the world with facts and rules.
 - $\text{fly}(X) \text{ :- bird}(X)$. It says If X is a bird, then it flies too.
 - $\text{lung_cancer}(X) \text{ :- smoking}(X)$. It says a person smoking has a lung cancer too.
- But in many cases,
 - Certain rules are hard to find, or difficult to express
- Instead, sometimes it is easier to assign probability to something or some event being True.

Probabilistic reasoning (PR)

- Many different factors can contribute to uncertainty,
 - Information gathered from unreliable sources
 - Error in data recording
 - Instrument Error
 - Etc.
- PR is useful when something is not certain, like ‘it will rain today’ or ‘Roger Federer will win the match’ or ‘Ali will pass the exam’.

Probability Theory

- Probability theory is the most popular way of representing uncertainty.
- It is a well-defined theory for representing and manipulating statements with uncertainty.
- The rule $\text{covid}(X) \text{ :- fever}(X)$ is replaced by $P(\text{covid} \mid \text{fever}) = 0.6$
- Facts (propositional statements) are represented via random variables with two or more values.
- Example: if *Pneumonia* is a random variable, then it can have two values: **True** and **False**
- Each value can be achieved with some probability:
 - $P(\text{Pneumonia} = \text{True}) = 0.001$
 - $P(\text{WBCcount} = \text{high}) = 0.005$

Axioms of Probability Theory

- All probabilities between 0 and 1

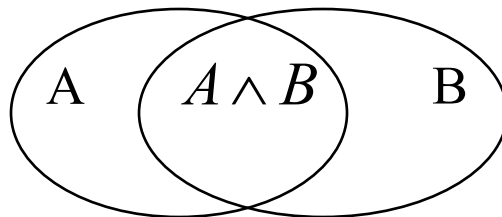
$$0 \leq P(A) \leq 1$$

- True proposition has probability 1, false has probability 0.

$$P(\text{true}) = 1 \quad P(\text{false}) = 0.$$

- The probability of disjunction is:

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B)$$



Random Variables

- A random variable is some aspect of the world about which we (may) have uncertainty
 - R = Is it raining?
 - T = Is it hot or cold?
 - D = How long will it take to drive to work?
 - L = Where is the ghost?
- We denote random variables with capital letters
- Random variables have domains
 - R in $\{\text{true}, \text{false}\}$ (often write as $\{+r, -r\}$)
 - T in $\{\text{hot}, \text{cold}\}$
 - D in $[0, \infty)$
 - L in possible locations, maybe $\{(0,0), (0,1), \dots\}$

Probability Distributions

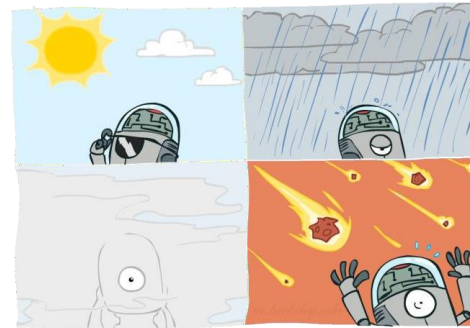
- Associate a probability with each outcome

- Temperature:


$$P(T)$$

T	P
hot	0.5
cold	0.5

- Weather:


$$P(W)$$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Distributions

- Unobserved random variables have distributions

$$P(T)$$

T	P
hot	0.5
cold	0.5

$$P(W)$$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Shorthand notation:

$$P(\textit{hot}) = P(T = \textit{hot}),$$

$$P(\textit{cold}) = P(T = \textit{cold}),$$

$$P(\textit{rain}) = P(W = \textit{rain}),$$

...

OK if all domain entries are unique

- A distribution is a TABLE of probabilities of values
- A probability (lower case value) is a single number

$$P(W = \textit{rain}) = 0.1$$

- Must have: $\forall x \ P(X = x) \geq 0$ and $\sum_x P(X = x) = 1$

Joint Distributions

- A *joint distribution* over a set of random variables: $X_1, X_2, \dots, X_n$ specifies a real number for each assignment (or *outcome*):

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey: $P(x_1, x_2, \dots, x_n) \geq 0$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Events

- An *event* is a set E of outcomes

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

- From a joint distribution, we can calculate the probability of any event

- Probability that it's hot AND sunny?
- Probability that it's hot?
- Probability that it's hot OR sunny?

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- Typically, the events we care about are *partial assignments*, like $P(T=\text{hot})$

Marginal Distributions

- Marginal distribution is the probability distribution of a single variable drawn from the joint distribution. Thus, it is a sub-table.
- Marginalization means value-wise sum of probabilities for a variable.

$P(T, W)$			$P(T)$	
T	W	P	T	P
hot	sun	0.4	hot	0.5
hot	rain	0.1	cold	0.5
cold	sun	0.2		
cold	rain	0.3		

$$P(t) = \sum_s P(t, s)$$
$$P(s) = \sum_t P(t, s)$$

$P(W)$	
W	P
sun	0.6
rain	0.4

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

Quiz: Marginal Distributions

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1



$$P(x) = \sum_y P(x, y)$$

$P(X)$

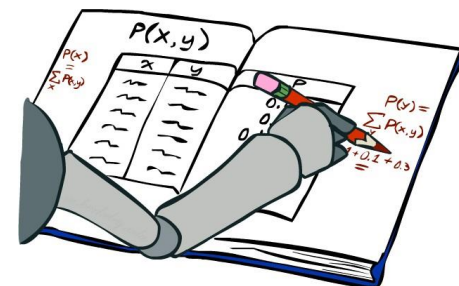
X	P
+x	
-x	



$$P(y) = \sum_x P(x, y)$$

$P(Y)$

Y	P
+y	
-y	



Quiz: Marginal Distributions

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

$$P(x) = \sum_y P(x, y)$$

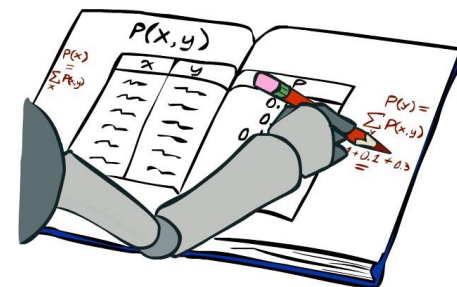
$$P(y) = \sum_x P(x, y)$$

$P(X)$

X	P
+x	.5
-x	.5

$P(Y)$

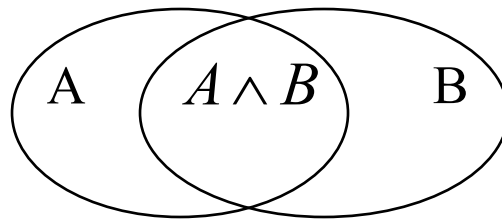
Y	P
+y	.6
-y	.4



Conditional Probability

- $P(A | B)$ is the probability of A given B
- Assumes that B is all and only information known.
- Defined by:

$$P(A | B) = \frac{P(A \wedge B)}{P(B)}$$



Quiz: Conditional Probabilities

- $P(+x \mid +y) ?$

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

- $P(-x \mid +y) ?$

- $P(-y \mid +x) ?$

Quiz: Conditional Probabilities

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

▪ $P(+x \mid +y) ?$

$.2/.6 = 1/3$

▪ $P(-x \mid +y) ?$

$.4/.6 = 2/3$

▪ $P(-y \mid +x) ?$

$.3/.5 = .6$

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W|T)$

$P(W T = \text{hot})$	
W	P
sun	
rain	

$P(W T = \text{cold})$	
W	P
sun	
rain	

Joint Distribution

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

Conditional Distributions

- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

$P(W T)$	$P(W T = \text{hot})$	
	W	P
	sun	0.8
	rain	0.2
	$P(W T = \text{cold})$	
	W	P
	sun	0.4
	rain	0.6

Joint Distribution

$P(T, W)$		
T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

The Product Rule

- Sometimes we have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y) \quad \longleftrightarrow \quad P(x|y) = \frac{P(x, y)}{P(y)}$$

The Product Rule

$$P(y)P(x|y) = P(x, y)$$

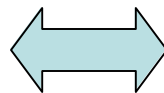
- Example:

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3



$P(D, W)$

D	W	P
wet	sun	
dry	sun	
wet	rain	
dry	rain	

The Chain Rule

- More generally, can always write any joint distribution as an incremental product of conditional distributions.

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$

Independence

- A and B are *independent* iff:

$$P(A \mid B) = P(A)$$

$$P(B \mid A) = P(B)$$

These two constraints are logically equivalent

- Therefore, if A and B are independent:

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)} = P(A)$$

$$P(A \wedge B) = P(A)P(B)$$

Example:

Independence?

$P_1(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$P(T)$

T	P
hot	0.5
cold	0.5

$P_2(T, W)$

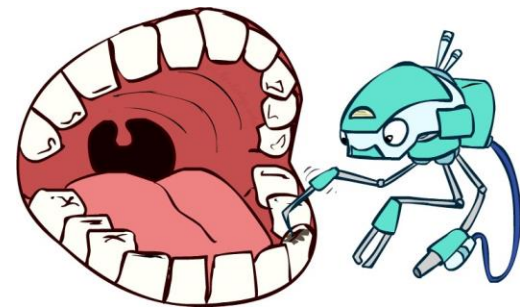
T	W	P
hot	sun	0.3
hot	rain	0.2
cold	sun	0.3
cold	rain	0.2

$P(W)$

W	P
sun	0.6
rain	0.4

Conditional Independence

- Unconditional (absolute) independence very rare (why?)
- $P(\text{Toothache}, \text{Cavity}, \text{Catch})$
- If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:
 - $P(+\text{catch} \mid +\text{toothache}, +\text{cavity}) = P(+\text{catch} \mid +\text{cavity})$
- The same independence holds if I don't have a cavity:
 - $P(+\text{catch} \mid +\text{toothache}, -\text{cavity}) = P(+\text{catch} \mid -\text{cavity})$
- Catch is *conditionally independent* of Toothache given Cavity:
 - $P(\text{Catch} \mid \text{Toothache}, \text{Cavity}) = P(\text{Catch} \mid \text{Cavity})$
- Equivalent statements:
 - $P(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity})$
 - $P(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = P(\text{Toothache} \mid \text{Cavity}) P(\text{Catch} \mid \text{Cavity})$
 - One can be derived from the other easily



Bayes' Rule

- A famous rule of probabilistic inference.
- Based on the formula of conditional probability, there are two ways to factor a joint distribution over two variables:

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Why is this at all helpful?
 - Note that typically the opposite conditional probabilities are given to us: they are much easier to estimate
 - Solution: apply Bayes rule to reverse the conditioning variable
 - Foundation of many systems

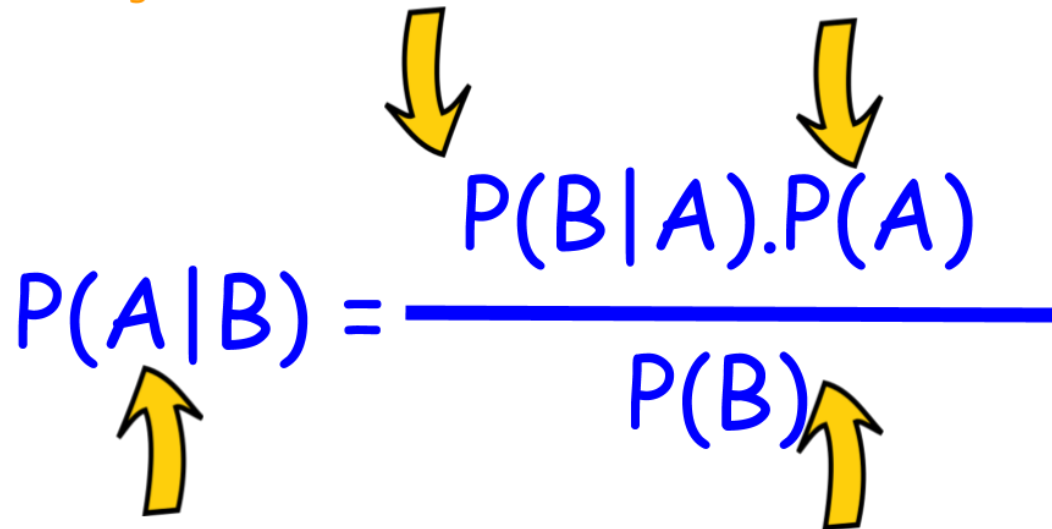
Bayes Rule (Theorem)

LIKELIHOOD

The probability of "B" being True, given "A" is True

PRIOR

The probability "A" being True. This is the knowledge.



The diagram shows the Bayes' Theorem formula with four yellow arrows pointing to its components: one from 'LIKELIHOOD' to $P(B|A)$, one from 'PRIOR' to $P(A)$, one from 'POSTERIOR' to $P(A|B)$, and one from 'MARGINALIZATION' to $P(B)$.

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)}$$

POSTERIOR

The probability of "A" being True, given "B" is True

MARGINALIZATION

The probability "B" being True.

Example of Bayes Theorem

- Given:
 - A doctor knows that meningitis causes stiff neck 50% of the time
 - Prior probability of any patient having meningitis is $1/50,000$
 - Prior probability of any patient having stiff neck is $1/20$
- If a patient has stiff neck, what's the probability he/she has meningitis?

$$P(M | S) = \frac{P(S | M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

Probabilistic Reasoning – Example

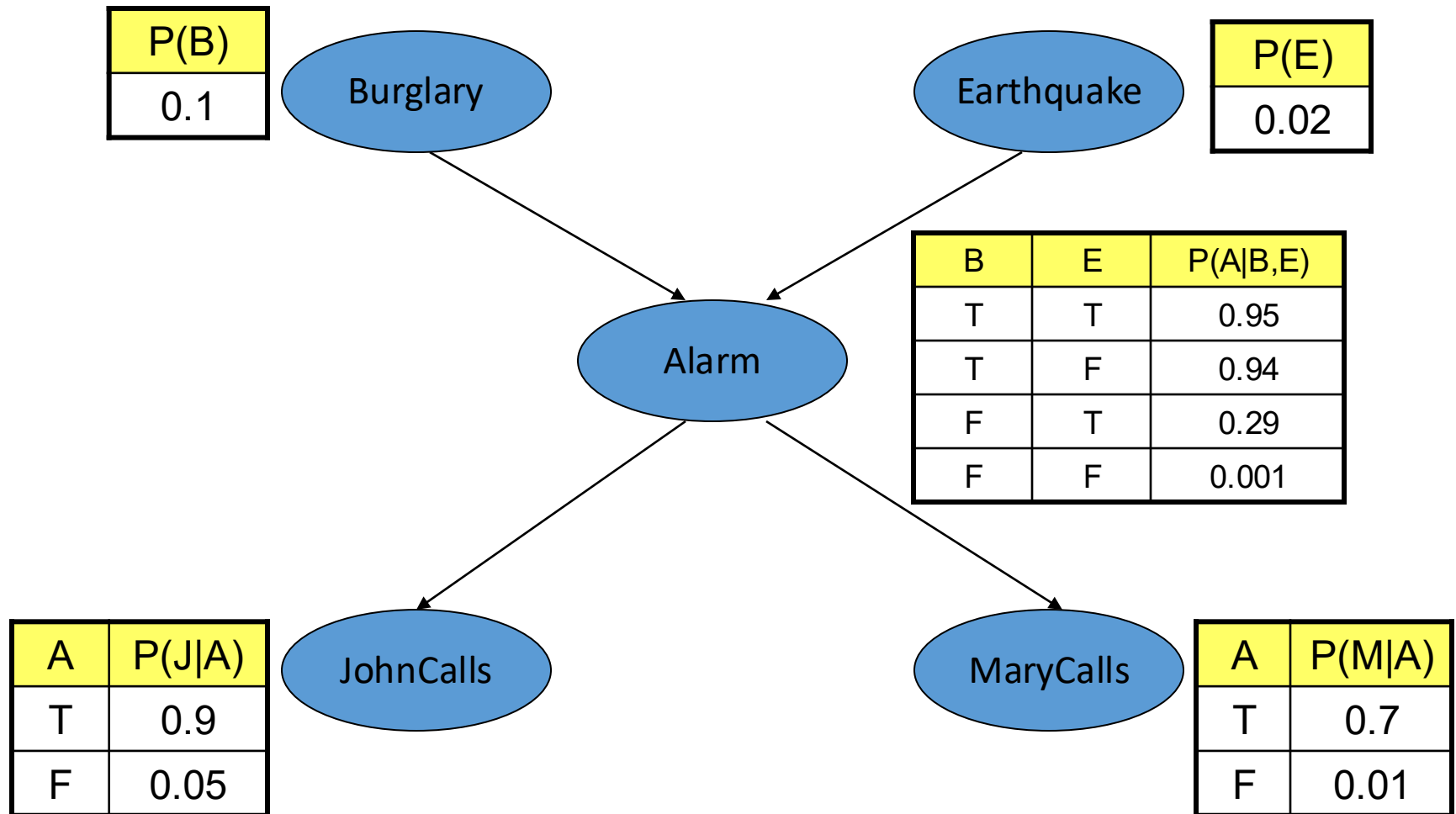
- Assume your house has an alarm system against burglary. You live in the seismically active area and the alarm system can get occasionally set off by an earthquake. You have two neighbors, Mary and John, who do not know each other. If they hear the alarm they call you, but this is not guaranteed.
- John always call when hears alarm, but confuses alarm with phone ringing (and calls then also)
- Mary likes loud music and sometimes misses alarm

Probabilistic Reasoning – Example

Cause and Effect Relationships

- Burglary $\rightarrow$ Alarm Goes Off
- Earthquake $\rightarrow$ Alarm Goes Off
- Alarm Goes Off $\rightarrow$ John Calls
- Alarm Goes Off $\rightarrow$ Mary Calls

Probabilistic Reasoning – Example



Inference Problem

- **Find the posterior probability distribution for some variable(s) given direct or virtual evidence about other variable(s)**
- Given evidence about who has and hasn't called, estimate the probability of a burglary.

Terminologies

- Joint Distribution
 - A joint probability gives us a full picture about how random variables are related
- Marginal Distribution
 - Marginalization lets us to focus one aspect of the picture
- Prior probability
 - Belief about a hypothesis h (or event) before obtaining observations
- Posterior probability
 - Belief about a hypothesis after obtaining observations

Solving using Joint Distribution

- A problem domain is modeled by a list of variables $X_1, X_2, \dots, X_n$.
- Knowledge about the problem domain is represented by a joint probability $P(X_1, X_2, \dots, X_n)$.
- General probability distribution of 5 variables with 2 states each has $2^5 = 32$ possible values and $2^5 - 1$ probabilities need to be specified.
- Problems
 - Exponential model size
 - Knowledge acquisition difficult
 - Exponential storage and inference

Solving using Joint Distribution

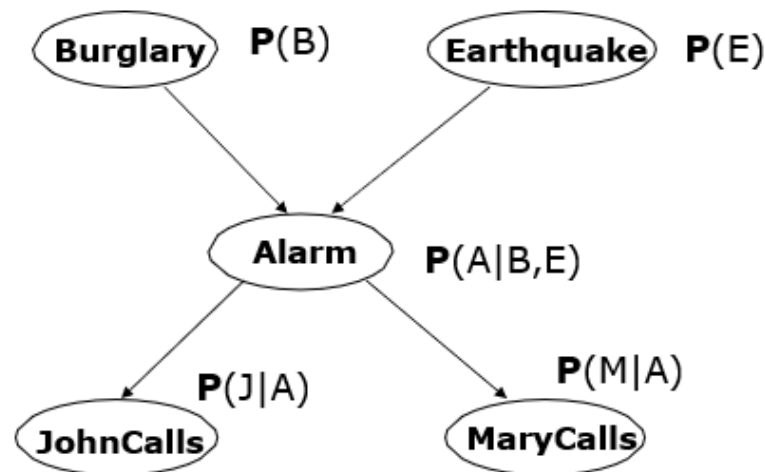
- Two problems with using full joint distributions for probabilistic models:
 1. Unless there are only a few variables, the joint is WAY too big to represent explicitly
 2. Hard to estimate anything empirically about more than a few variables at a time

Bayesian Network

- Technique for describing complex joint distributions using a bunch of simple, local distributions
 - We describe how variables locally interact
 - Local interactions chain together to give global, indirect interactions
- Graphical representation of the joint distribution.
- Give solutions to the data acquisition bottlenecks.

Bayesian Network

- A BN is a Directed Acyclic Graph (DAG) in which:
- A set of random variables makes up the nodes in the network.
- A set of directed links or arrows connects pairs of nodes.
- (Missing) links encode independences.
- Each node has a conditional probability table that quantifies the effects the parents have on the node.



Bayesian Network

- Assumes that each node is conditionally independent of all its non-descendants given its parents.
- E.g., JohnCalls is independent of Burglary, Earthquake, and MaryCalls given the value of Alarm.
- A node is conditionally independent of all other nodes in the network, given its parents, children, and children's parents — that is, given its **Markov blanket**.
- For example, Burglary is independent of JohnCalls and MaryCalls, given Alarm and Earthquake.

Bayesian Network

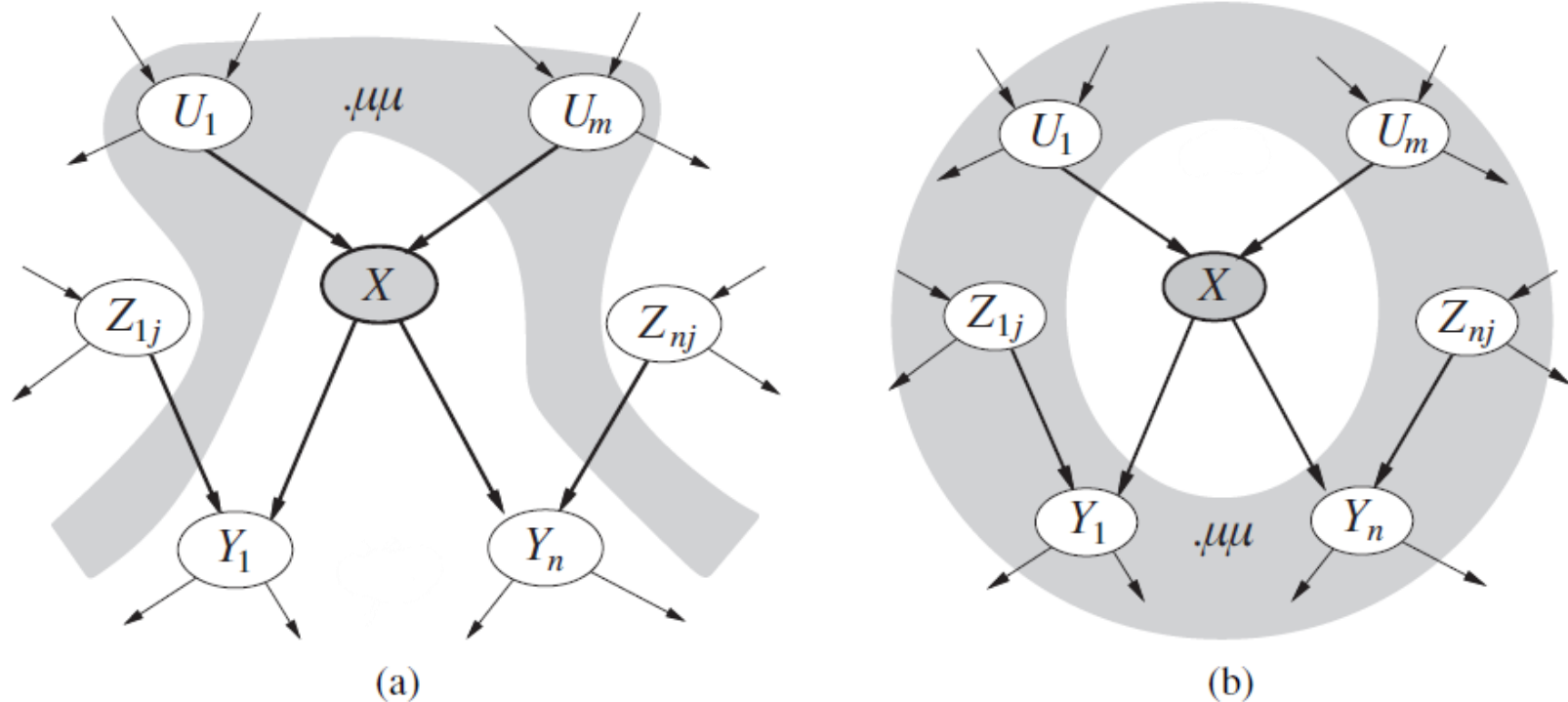
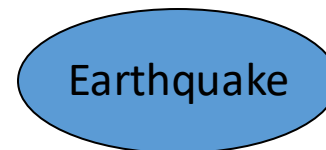


Figure 14.4 (a) A node X is conditionally independent of its non-descendants (e.g., the Z_{ij} s) given its parents (the U_i s shown in the gray area). (b) A node X is conditionally independent of all other nodes in the network given its Markov blanket (the gray area).

Bayesian Network – Example

P(B)
0.1



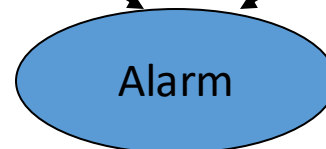
P(E)
0.02

of parameters of the BBN:

$$2^3 + 2(2^2) + 2(2) = 20$$

One parameter in every conditional is for free:

$$2^2 + 2(2) + 2(1) = 10$$



B	E	P(A B,E)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

A	P(J A)
T	0.9
F	0.05



A	P(M A)
T	0.7
F	0.01

A Bayes net encodes the full joint distribution (FJPD), often with far less parameters

Full joint distribution in BNs

Product of all conditional probabilities is the joint probability of all variables.

Full joint distribution is defined in terms of local conditional distributions (obtained via the chain rule):

$$\mathbf{P}(X_1, X_2, \dots, X_n) = \prod_{i=1, \dots, n} \mathbf{P}(X_i \mid pa(X_i))$$

Example:

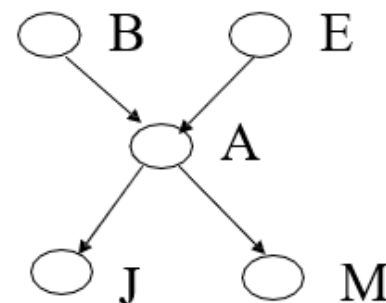
Assume the following assignment of values to random variables

$$B=T, E=T, A=T, J=T, M=F$$

Then its probability is:

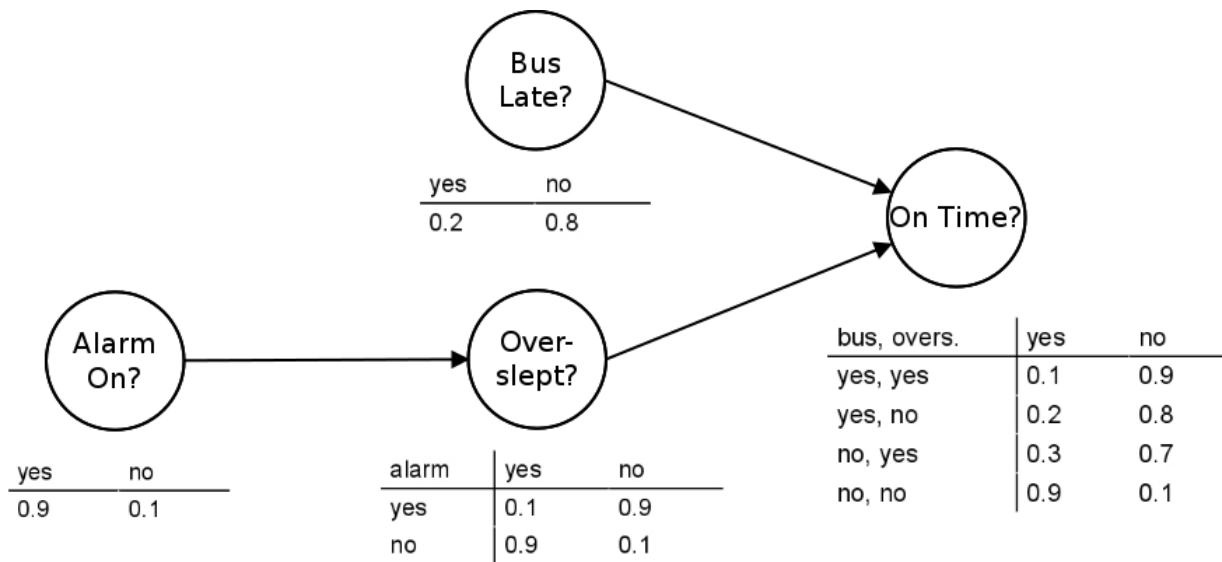
$$P(B=T, E=T, A=T, J=T, M=F) =$$

$$P(B=T)P(E=T)P(A=T \mid B=T, E=T)P(J=T \mid A=T)P(M=F \mid A=T)$$



Full joint distribution in BNs (Practice)

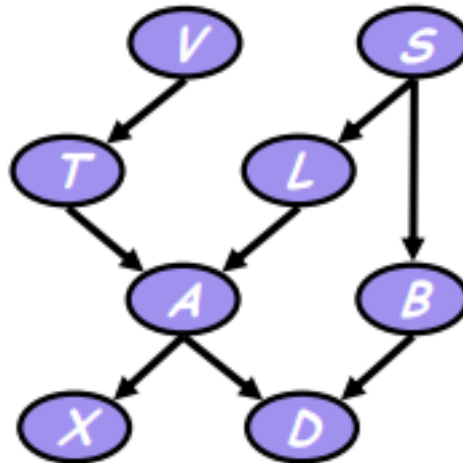
Write the joint probability distribution for this Bayesian Network



$$P(AO, OS, BL, OT) = P(AO) P(BL) P(OS/AO) P(OT/OS, BL)$$

Full joint distribution in BNs (Practice)

Write the joint probability distribution for this Bayesian Network



$$P(v, s, t, l, a, b, x, d) =$$

$$P(v)P(s)P(t|v)P(l|s)P(b|s)P(a|t,l)P(x|a)P(d|a,b)$$

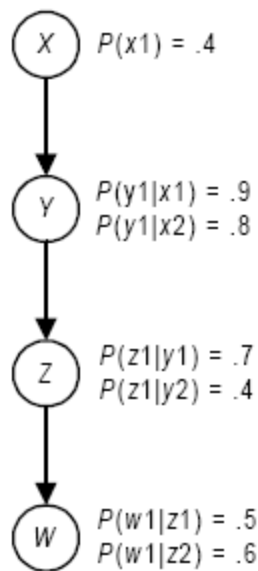
Bayesian Network

- The intuitive meaning of an arrow from a parent to a child is that the parent directly influences the child.
- The direction of this influence is often taken to represent causal influence.
- These influences are quantified by conditional probabilities.
- Arrows do not necessarily represent causal relations; they only encode conditional independencies.

Bayesian Network

- Solving simple Inference Problem using a single connected BN.
- ***Find the posterior probability distribution for some variable(s) given direct or virtual evidence about other variable(s)***

Inference in Singly Connected Network



(a)

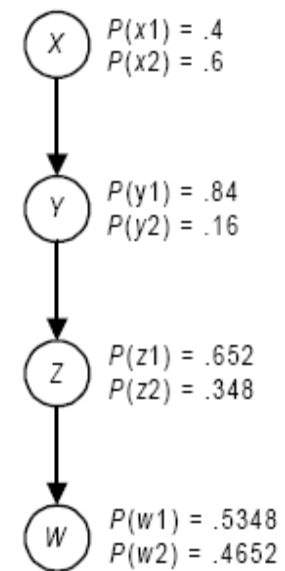
$$P(y1) = P(y1 | x1) P(x1) + P(y1 | x2) P(x2)$$

$$P(y2) = 1 - P(y1)$$

$$P(z1) = P(z1 | y1) P(y1) + P(z1 | y2) P(y2)$$

$$P(z2) = 1 - P(z1)$$

$$P(w1) = ???$$



(b)

We have to find marginal of $w1$, which is

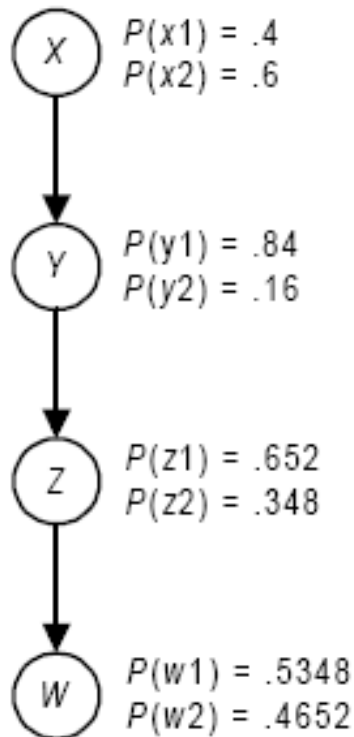
$$P(W=w1) = P(W=w1, Z=z1) + P(W=w1, Z=z2)$$

$$P(W=w1) = P(W=w1 | Z=z1) P(Z=z1) + P(W=w1 | Z=z2) P(Z=z2)$$

For this, we must first find $P(Z=z1)$ and $P(Z=z2)$.

Before this, we must find $P(Y=y1)$ and $P(Y=y2)$

Inference in Singly Connected Networks



Suppose we get evidence that w1 is true, i.e., $P(w1) = 1$.

Now compute the posterior probabilities:

$P^*(z1)$, $P^*(y1)$, $P^*(x1)$

$$P^*(z1) = P(z1 \mid w1) P(w1) \quad \leftarrow P(w1)=1$$
$$+ P(z1 \mid w2) P(w2) \quad \leftarrow P(w2)=0$$

Computing $P(z1 \mid w1)$ using Bayes rule:

$$P(z1 \mid w1) = P(w1 \mid z1) P(z1) / P(w1)$$

$$P(z1 \mid w1) = 0.5 \times 0.652 / 0.5348 = 0.61$$

=>

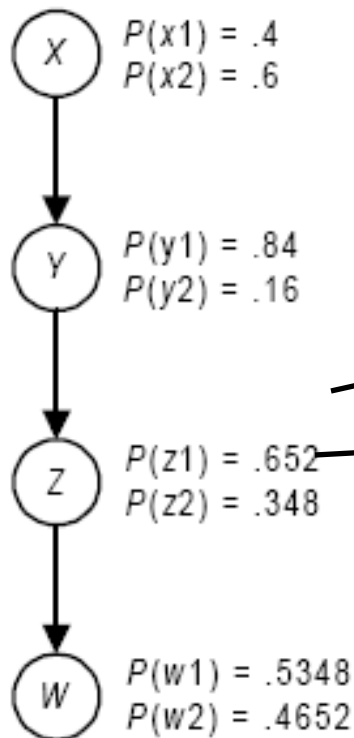
$$P^*(z1) = 0.61 \times 1 + 0 = 0.61$$

So,

$$P^*(z2) = 1 - 0.61 = 0.39$$

Computation of $P^*(y1)$
on the next slide

Inference in Singly Connected Networks



Now update the probability of Y.

$$P^*(y1) = P(y1 | z1) P(z1) + P(y1 | z2) P(z2)$$

$$P(z1) = 0.61$$

$$P(z2) = 0.39$$

} Computed on the previous slide

Computing $P(y1 | z1)$ and $P(y1 | z2)$ using Bayes theorem:

$$P(y1 | z1) = P(z1 | y1) P(y1) / P(z1)$$

$$P(y1 | z1) = 0.7 \times 0.84 / 0.652 = 0.90$$

$$P(y1 | z2) = P(z2 | y1) P(y1) / P(z2)$$

$$P(y1 | z2) = 0.3 \times 0.84 / 0.348 = 0.72$$

Finally plug the values in the equation at the top

$$P^*(y1) = P(y1 | z1) P(z1) + P(y1 | z2) P(z2)$$

$$= 0.90 \times 0.61 + 0.72 \times 0.39$$

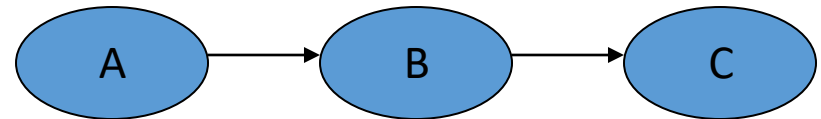
$$= 0.83$$

Confirm the results using Genie.

Finally compute $P^*(x1)$

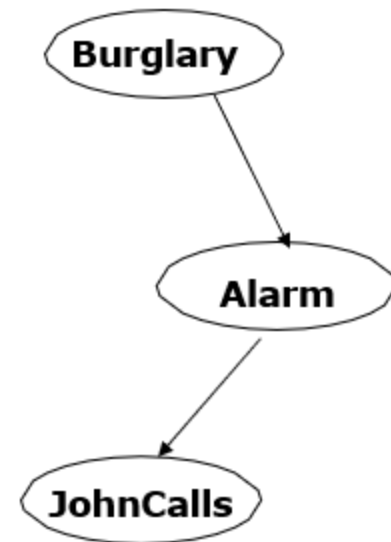
Conditional Dependence/Independence

Causal Chains



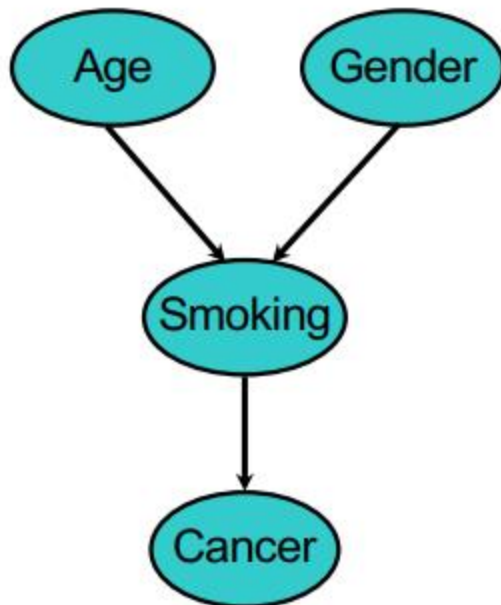
- Smoking causes cancer which causes dyspnoea
- If we have evidence on B then A and C become independent.
- If we known Alarm has set off or not, then if John calls or not is independent of whether the burglary happened or not.
- Knowing that the Alarm went off means that also knowing that a Burglary has taken place will not increase my belief that Mary Calls

$$P(J \mid A, B) = P(J \mid A)$$
$$P(J, B \mid A) = P(J \mid A)P(B \mid A)$$



Conditional Dependence/Independence

Causal Chains



Cancer is conditionally independent of *Age* and *Gender* given *Smoking*

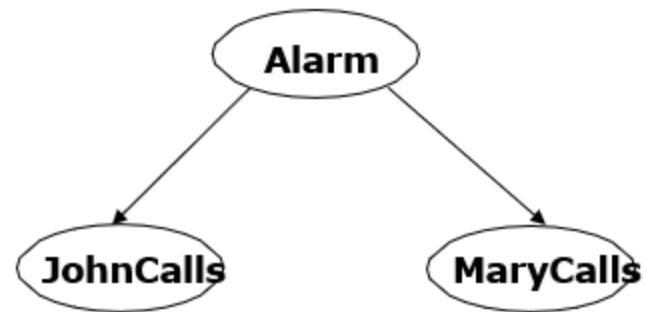
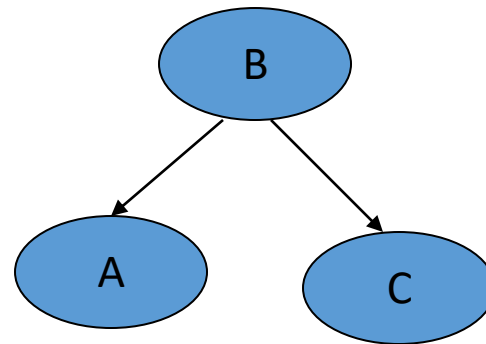
Conditional Dependence/Independence

Common Causes

- Cold is a common cause of sneezing and fever.
- If we have evidence on B then A and C become independent, otherwise they are dependent.
- MaryCalls is independent of JohnCalls given Alarm

$$P(J \mid A, M) = P(J \mid A)$$

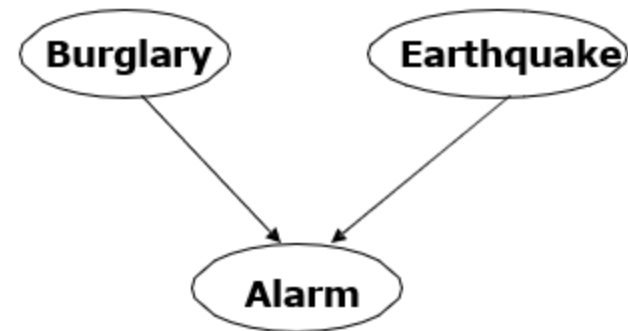
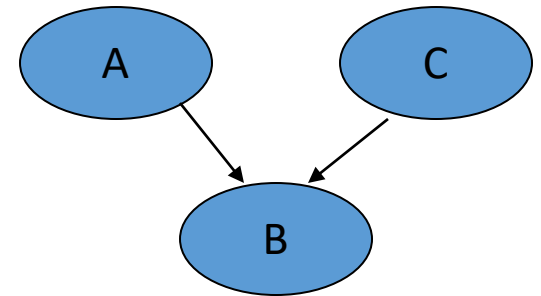
$$P(J, M \mid A) = P(J \mid A)P(M \mid A)$$



Conditional Dependence/Independence

Common Effects

- Rain and law and order problems are common causes of traffic jams
- If we have evidence on B then A and C become dependent.
- Burglary and Earthquake become dependent given Alarm!
- If I already know that the Alarm went off, my further knowing that there has been an Earthquake, decreases my belief that there has been a Burglary. Called “explaining away.”
- Similarly, if B has descendant D and D is given, then A and C are not independent.



Inference Using Joint Distributions in BN

- Having a BN, any joint probability can be computed as

$$P(x_1, \dots, x_N) = \prod_i P(x_i \mid \text{parents}(x_i))$$

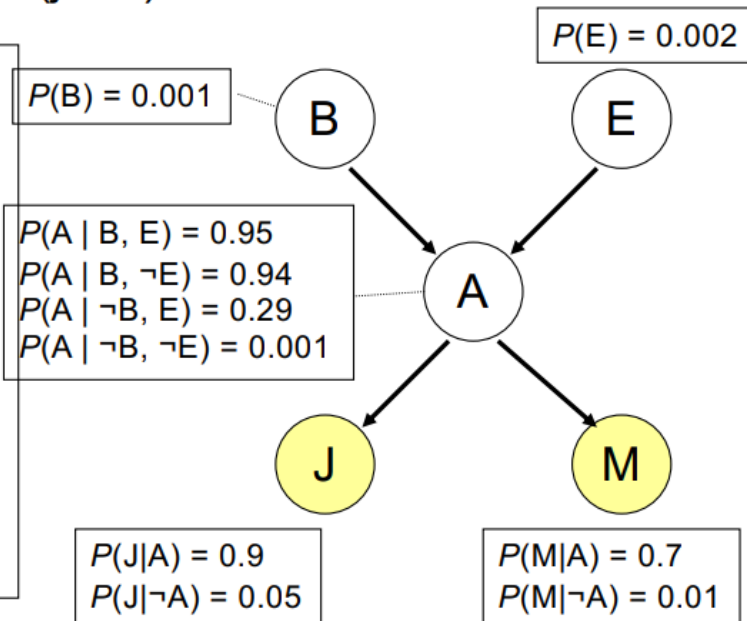
- With this joint distribution, we can compute any conditional probability, $P(Q \mid E)$; thus we can perform any inference.
- How?

Inference by Enumeration

$$P(Q \mid E) = \frac{\sum_{\text{joint matching } Q \text{ AND } E} P(\text{joint})}{\sum_{\text{joint matching } E} P(\text{joint})} \quad \text{by def. of cond. prob.}$$

For example: $P(B \mid J, \neg M)$

1. Compute $P(B, J, \neg M)$
2. Compute $P(J, \neg M)$
3. Return $P(B, J, \neg M) / P(J, \neg M)$



Inference by Enumeration

Inference by Enumeration

$$P(Q | E) = \frac{\sum_{\text{joint matching Q}} P(\text{joint matching Q} | E)}{\sum_{\text{joint matching}} P(\text{joint matching} | E)}$$

For example: $P(B | J, \neg M)$

1. Compute $P(B, J, \neg M)$
2. Compute $P(J, \neg M)$
3. Return $P(B, J, \neg M) / P(J, \neg M)$

Compute the joints (4 of them)

$$\begin{aligned} &P(B, J, \neg M, \textcolor{red}{A}, \textcolor{red}{E}) \\ &P(B, J, \neg M, \textcolor{red}{A}, \neg \textcolor{red}{E}) \\ &P(B, J, \neg M, \neg \textcolor{red}{A}, \textcolor{red}{E}) \\ &P(B, J, \neg M, \neg \textcolor{red}{A}, \neg \textcolor{red}{E}) \end{aligned}$$

Each is $O(N)$ for sparse graph

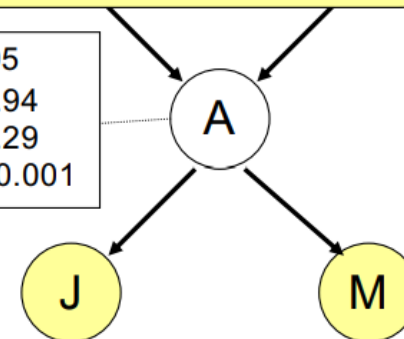
$$P(x_1, \dots, x_N) = \prod_i P(x_i | \text{parents}(x_i))$$

Sum them up

$$\begin{aligned} P(A | B, E) &= 0.95 \\ P(A | B, \neg E) &= 0.94 \\ P(A | \neg B, E) &= 0.29 \\ P(A | \neg B, \neg E) &= 0.001 \end{aligned}$$

$$\begin{aligned} P(J|A) &= 0.9 \\ P(J|\neg A) &= 0.05 \end{aligned}$$

$$\begin{aligned} P(M|A) &= 0.7 \\ P(M|\neg A) &= 0.01 \end{aligned}$$



Inference by Enumeration

Inference by Enumeration

$$P(Q \mid E) = \frac{\sum_{\text{joint matching Q}} P(\text{joint matching Q} \mid E)}{\sum_{\text{joint matching}} P(\text{joint matching})}$$

For example: $P(B \mid J, \neg M)$

1. Compute $P(B, J, \neg M)$
2. Compute $P(J, \neg M)$
3. Return $P(B, J, \neg M) / P(J, \neg M)$

Compute the joints (8 of them)

$P(J, \neg M, B, A, E)$
 $P(J, \neg M, B, A, \neg E)$
 $P(J, \neg M, B, \neg A, E)$
 $P(J, \neg M, B, \neg A, \neg E)$
 $P(J, \neg M, \neg B, A, E)$
 $P(J, \neg M, \neg B, A, \neg E)$
 $P(J, \neg M, \neg B, \neg A, E)$
 $P(J, \neg M, \neg B, \neg A, \neg E)$

Each is $O(N)$ for sparse graph

$$P(x_1, \dots, x_N) = \prod_i P(x_i \mid \text{parents}(x_i))$$

Sum them up

Inference by Enumeration

$$P(Q | E) = \frac{\sum_{\text{joint matching Q AND E}} P(\text{joint})}{\sum_{\text{joint matching E}} P(\text{joint})}$$

Sum up 4 joints

Sum up 8 joints

= 0.002

For example: $P(B | J, \neg M)$

1. Compute $P(B, J, \neg M)$
2. Compute $P(J, \neg M)$
3. Return $P(B, J, \neg M) / P(J, \neg M)$

In general, if there are N Boolean variables, j query vars, and k evidence vars, how many joints to sum up?

Ans: K^{N-j}

$P(J|A) = 0.9$
 $P(J|\neg A) = 0.05$

$P(M|A) = 0.7$
 $P(M|\neg A) = 0.01$

Inference by Enumeration (Practice)

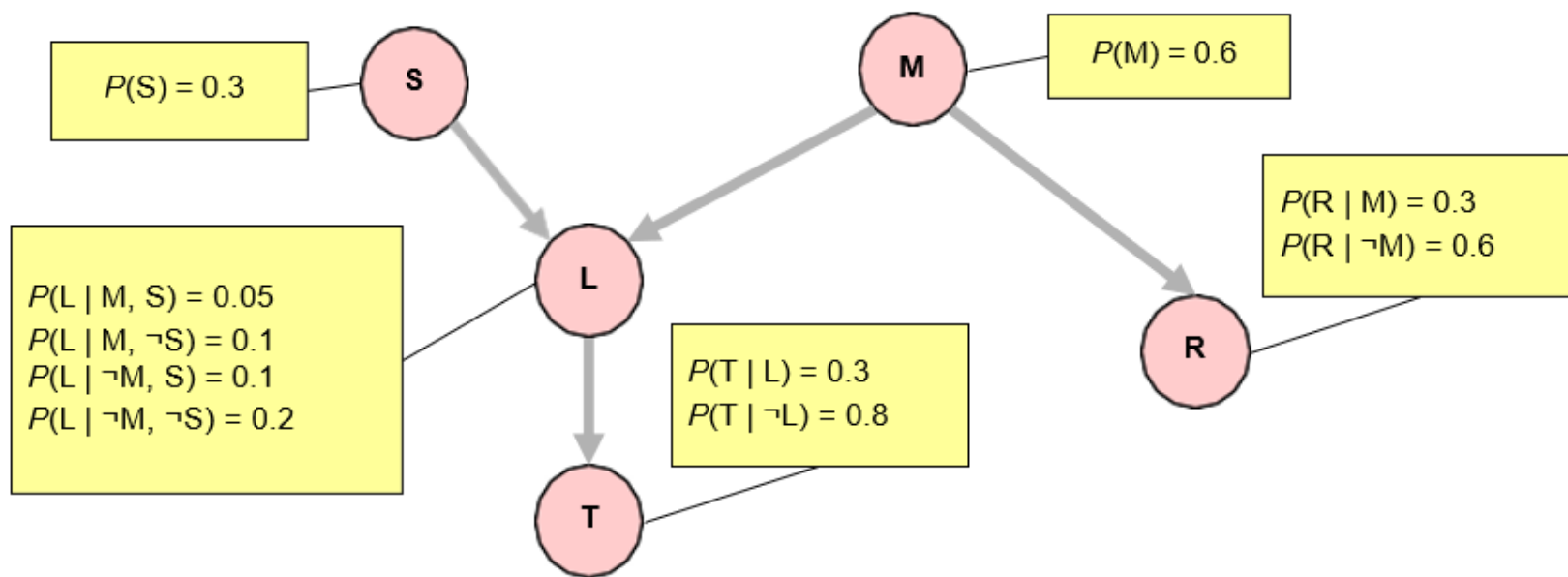
- $P(W)?$
- $P(W \mid \text{winter})?$
- $P(W \mid \text{winter, hot})?$

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Given: joint probability
distribution for a BN.

Another Example

Compute $P(R \mid T, \neg S)$ from the following Bayes Net

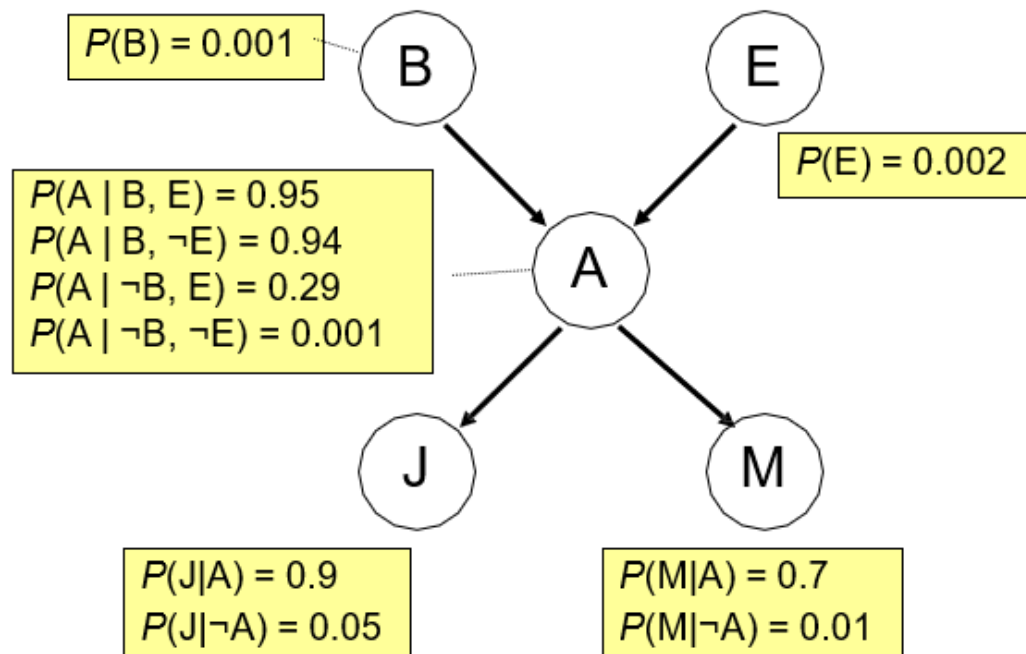


Inference By enumeration – Limitation

- In general, if there are N variables and evidence contains j variables, and each variable has k values, how many joints to sum up? k^{N-j}
- It is this summation that makes inference by enumeration inefficient
 - Computing conditional probabilities by enumerating all matching entries in the joint is expensive: Exponential in the number of variables
- Exact inference with an arbitrary Bayes Net is NP-Complete

Parameter (CPT) Learning for BN

- Where do you get these CPT numbers?
 - Ask domain experts, or
 - Learn from data



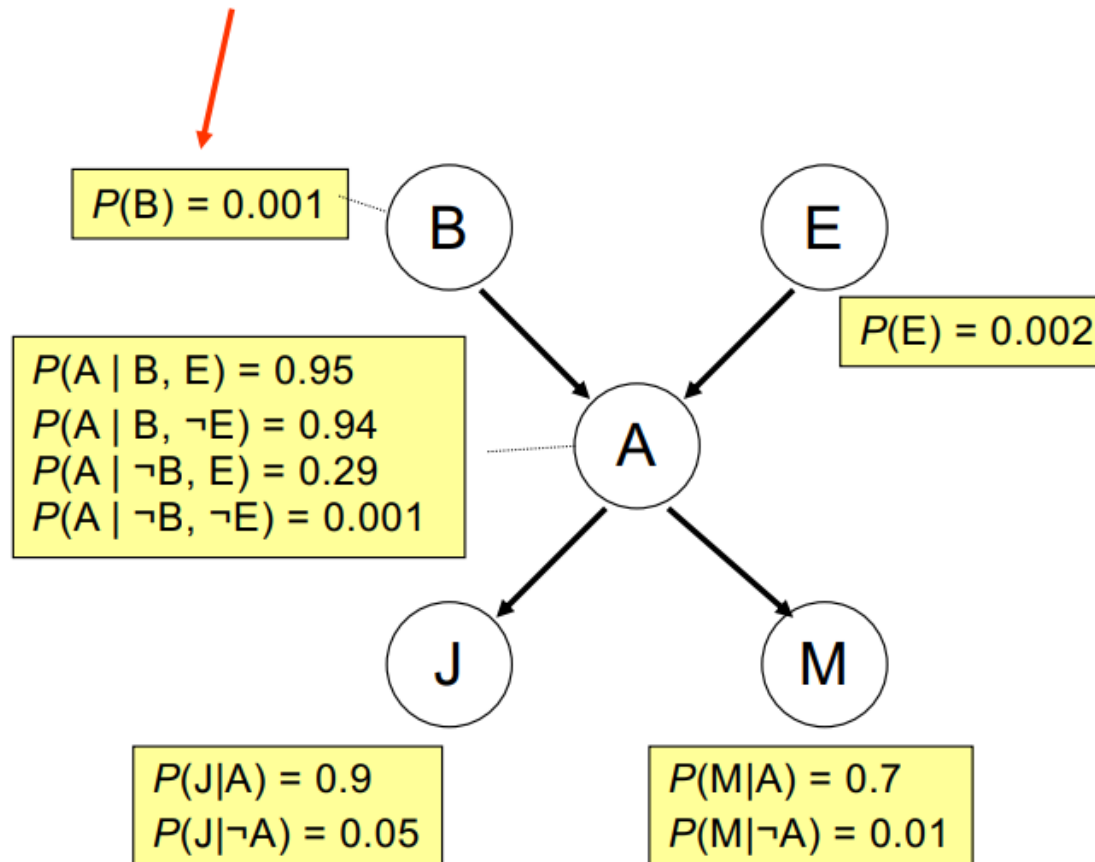
Parameter (CPT) Learning for BN

- Learn from a data set like this:

($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
(B, $\neg E$, A, J, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, E, A, J, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
(B, E, A, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

...

How to learn this CPT?



Parameter (CPT) Learning for BN

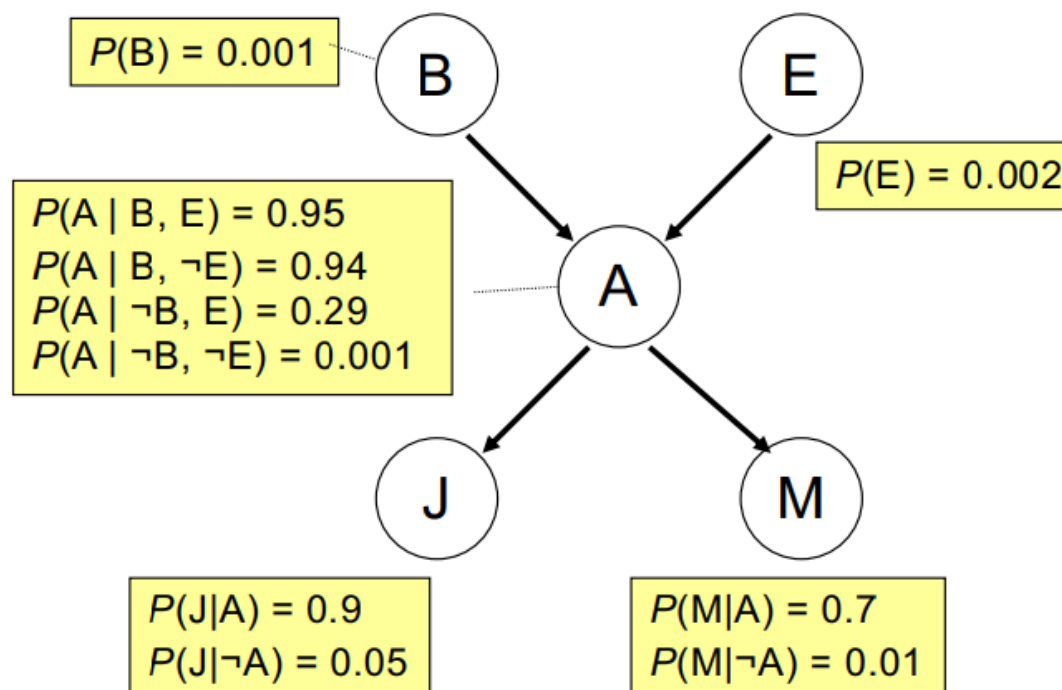
- Learn from a data set like this:

($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
(B, $\neg E$, A, J, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, E, A, J, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
(B, E, A, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

...

Count $\#(B)$ and $\#(\neg B)$ in dataset.

$$P(B) = \#(B) / [\#(B) + \#(\neg B)]$$



Parameter (CPT) Learning for BN

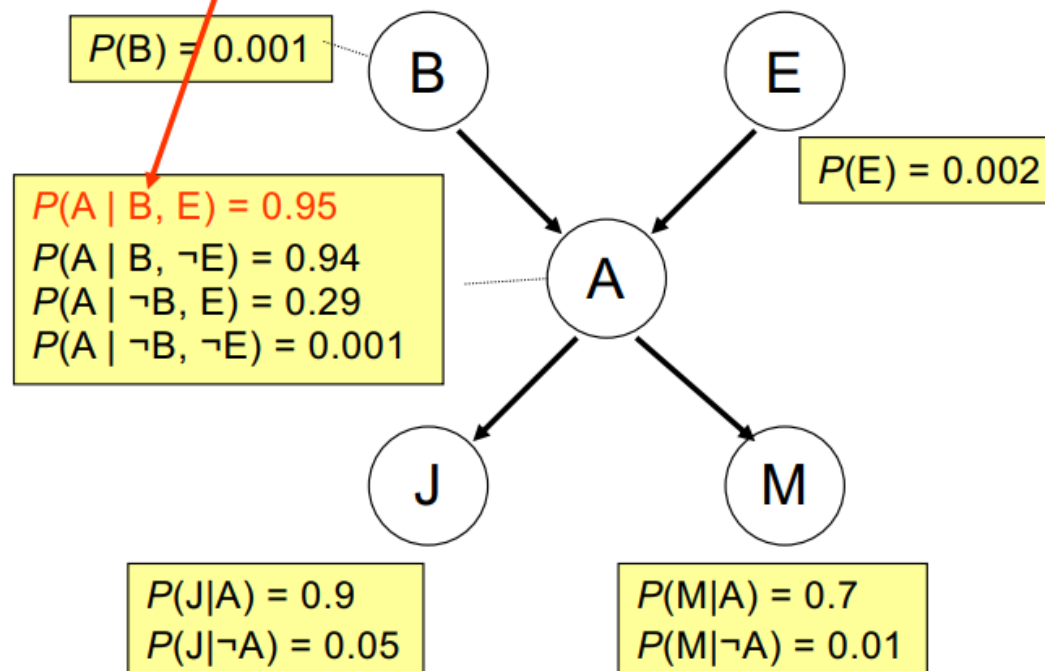
- Learn from a data set like this:

$(\neg B, \neg E, \neg A, J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(B, \neg E, A, J, M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, J, \neg M)$
 $(\neg B, E, A, J, M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$
 $(B, E, A, \neg J, M)$
 $(\neg B, \neg E, \neg A, \neg J, \neg M)$

...

Count $\#(A)$ and $\#(\neg A)$ in dataset where $B=\text{true}$ and $E=\text{true}$.

$$P(A|B,E) = \#(A) / [\#(A) + \#(\neg A)]$$



Parameter (CPT) Learning for BN

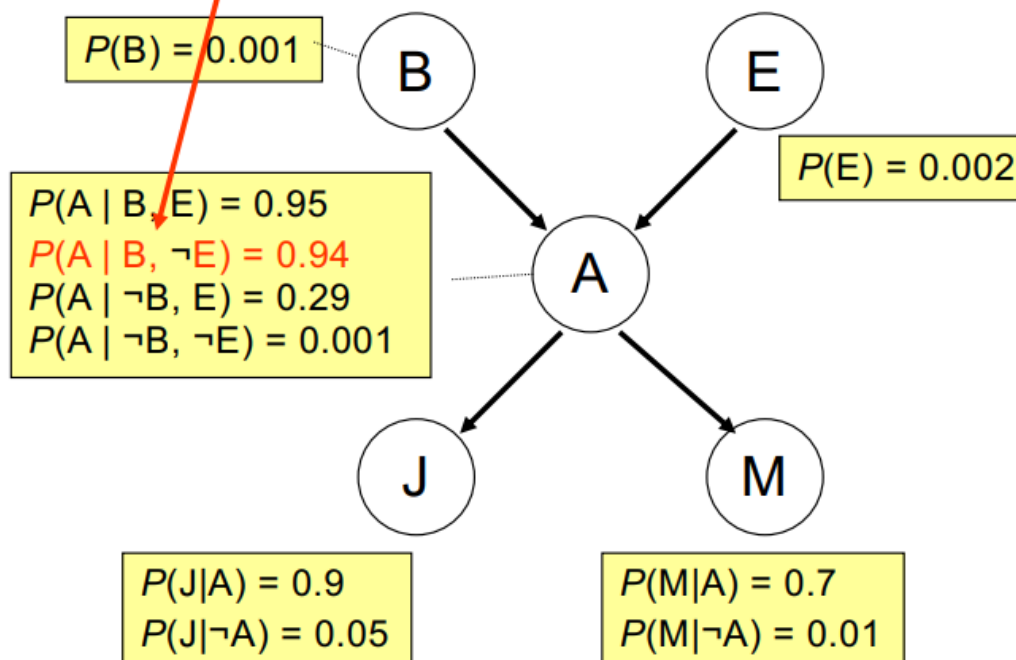
- Learn from a data set like this:

($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
(B, $\neg E$, A, J, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)
($\neg B$, E, A, J, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)
(B, E, A, $\neg J$, M)
($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

...

Count $\#(A)$ and $\#(\neg A)$ in dataset
where **B=true** and **E=false**.

$$P(A|B, \neg E) = \#(A) / [\#(A) + \#(\neg A)]$$



Parameter (CPT) Learning for BN

- 'Unseen event' problem

($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)

($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)

(B, $\neg E$, A, J, M)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, J, $\neg M$)

($\neg B$, E, A, J, M)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, M)

(B, E, A, $\neg J$, M)

($\neg B$, $\neg E$, $\neg A$, $\neg J$, $\neg M$)

...

Count $\#(A)$ and $\#(\neg A)$ in dataset where **B=true and E=true**.

$$P(A|B,E) = \#(A) / [\#(A) + \#(\neg A)]$$

What if there's **no** row with (B, E, $\neg A$, *, *) in the dataset?

Do you want to set

$$P(A|B,E) = 1$$

$$P(\neg A|B,E) = 0?$$

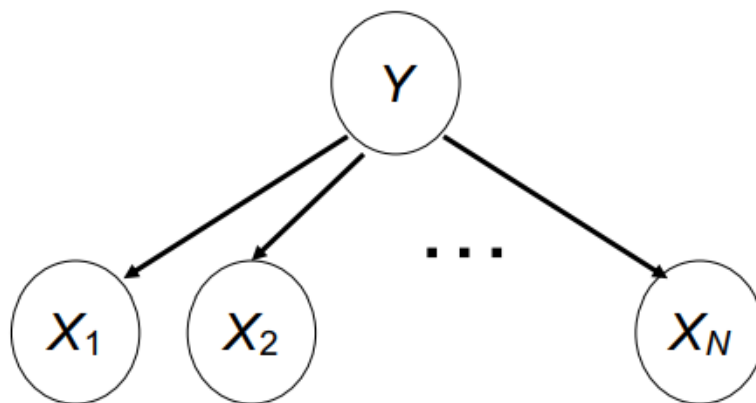
Why or why not?

Smoothing CPTs

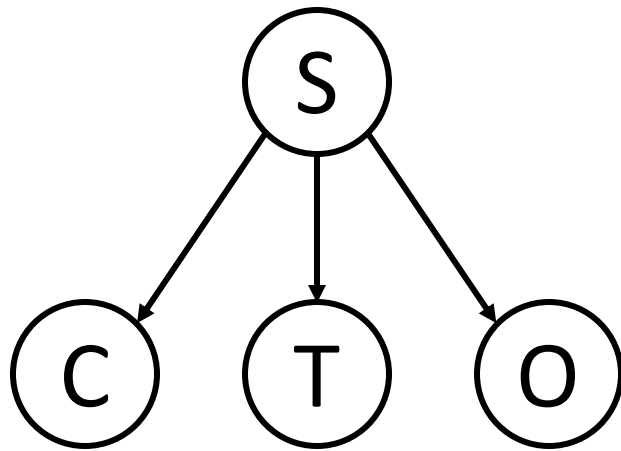
- “Add-1” smoothing: **add 1 to all counts**
- In the previous example, count $\#(A)$ and $\#(\neg A)$ in dataset where $B=\text{true}$ and $E=\text{true}$
 - $P(A|B,E) = [\#(A)+1] / [\#(A)+1 + \#(\neg A)+1]$
 - If $\#(A)=1$, $\#(\neg A)=0$:
 - without smoothing $P(A|B,E) = 1$, $P(\neg A|B,E) = 0$
 - with smoothing $P(A|B,E) = 0.67$, $P(\neg A|B,E) = 0.33$
 - If $\#(A)=100$, $\#(\neg A)=0$:
 - without smoothing $P(A|B,E) = 1$, $P(\neg A|B,E) = 0$
 - with smoothing $P(A|B,E) = 0.99$, $P(\neg A|B,E) = 0.01$
- Smoothing saves you when you don't have enough data, and hides away when you do

Bayesian Networks as Classifiers - Naïve Bayes

- **Naïve Bayes** classifier is a supervised learning algorithm which is a special Bayes Net structure. It has
 - a 'class' variable Y at root, compute $P(Y | X_1, \dots, X_N)$
 - evidence nodes X_i (observed features) are all leaves
 - **conditional independence between all evidence** assumed. Usually not valid, but often empirically OK



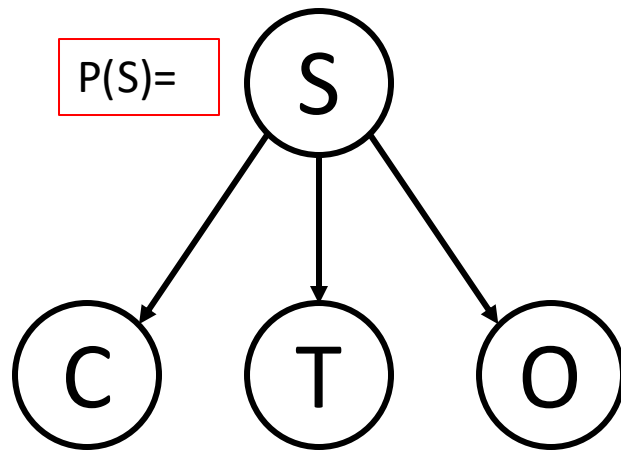
Naïve Bayes Classifier Example



Var	Meaning	Values
S	Stolen	Yes, No
C	Color	Red, Yellow
T	Type	Sports, SUV
O	Origin	Domestic, Imported

- Root is the class variable
- All the feature variables become effects of the class variable.
- What is stored in the CPTs?

Naïve Bayes Classifier Example



$$P(S)=$$

Var	Meaning	Values
S	Stolen	Yes, No
C	Color	Red, Yellow
T	Type	Sports, SUV
O	Origin	Domestic, Imported

$$P(C|S)=$$
$$P(C|\sim S)=$$

$$P(T|S)=$$
$$P(T|\sim S)=$$

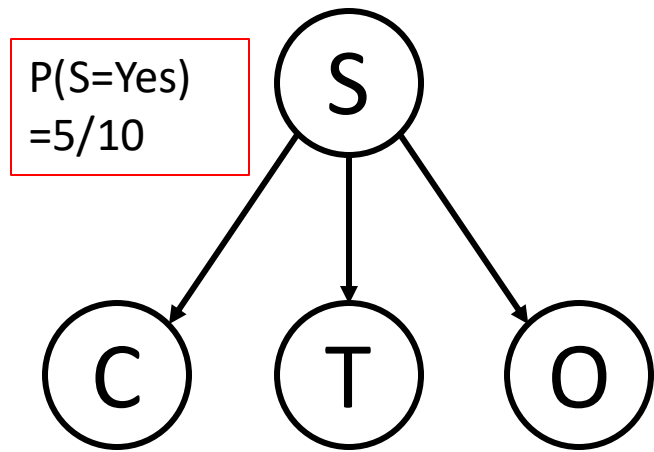
$$P(O|S)=$$
$$P(O|\sim S)=$$

- How do we get the probabilities?
- From labelled data (supervised data).

Naïve Bayes Classifier Example

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Naïve Bayes Classifier Example



$$P(S=\text{Yes}) = 5/10$$

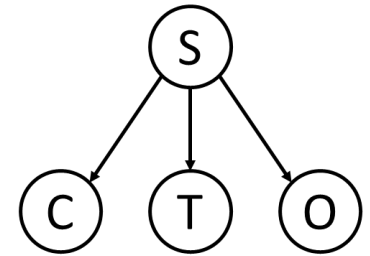
$$P(C=\text{Red} | S=\text{Yes}) = 3/5$$
$$P(C=\text{Red} | S=\text{No}) = 2/5$$

$$P(T=\text{Sports} | S=\text{Yes}) = 4/5$$
$$P(T=\text{Sports} | S=\text{No}) = 2/5$$

$$P(O=\text{Domestic} | S=\text{Yes}) = ??$$
$$P(O=\text{Domestic} | S=\text{No}) = ??$$

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Naïve Bayes Classifier Example

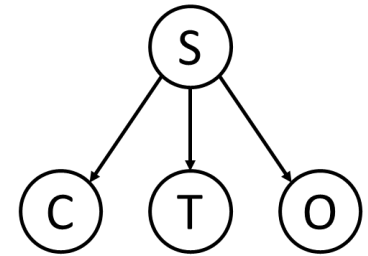


- To classify a test data, find the probability $P(S \mid C, T, O)$
- Test case: C=Red, T=SUV, O=Domestic
- Find $P(S=\text{Yes} \mid X)$ and $P(S=\text{No} \mid X)$. Where X are all the features as given in the test case.

$$P(S=\text{Yes} \mid C=\text{Red}, T=\text{SUV}, O=\text{Domestic})$$

$$= \frac{P(S=\text{Yes}) * P(C=\text{Red} \mid S=\text{Yes}) * P(T=\text{SUV} \mid S=\text{Yes}) * P(O=\text{Domestic} \mid S=\text{Yes})}{P(C=\text{Red}) * P(T=\text{SUV}) * P(O=\text{Domestic})}$$

Naïve Bayes Classifier Example



- Similarly find probability

$$P(S=\text{No} \mid C=\text{Red}, T=\text{SUV}, O=\text{Domestic})$$

$$= \frac{P(S=\text{No}) * P(C=\text{Red} \mid S=\text{No}) * P(T=\text{SUV} \mid S=\text{No}) * P(O=\text{Domestic} \mid S=\text{No})}{P(C=\text{Red}) * P(T=\text{SUV}) * P(O=\text{Domestic})}$$

- Then, the class label with the higher probability becomes the predicted class label.

Applications

- Medical diagnosis systems
- Manufacturing system diagnosis
- Computer systems diagnosis
- Network systems diagnosis
- Helpdesk troubleshooting
- Information retrieval
- Customer modeling

Applications

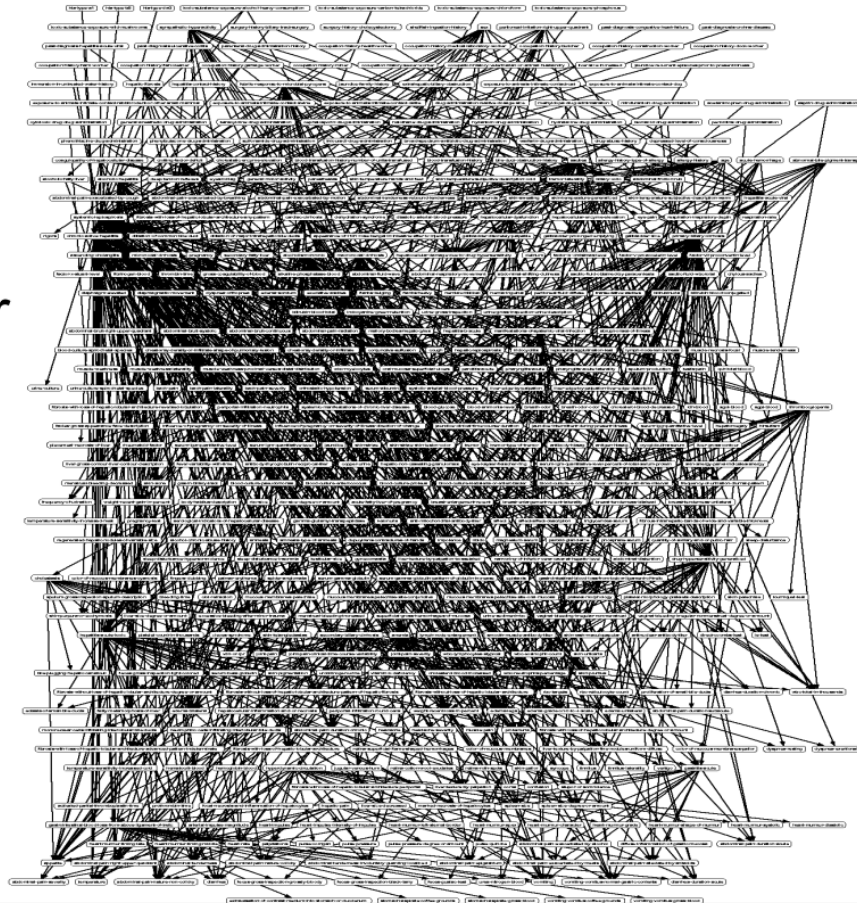
Pathfinder

- Pathfinder was one of the first BN systems
- It performed diagnosis of lymph-node diseases
- It dealt with over 60 diseases and 100 symptoms and test results
- 14,000 probabilities
- Commercialized and applied to about 20 tissue types

Applications

Pathfinder
Bayes
Net

448 nodes,
906 arcs



Resources

- Some slides were adopted from
<https://www.cs.utexas.edu/~mooney/cs343/>
https://pages.cs.wisc.edu/~dyer/cs540/notes/13_bayes-net.pdf
- slides adapted from Dan Klein, Pieter Abbeel ai.berkeley.edu And Dan Weld, Luke Zettlemoyer

Video Links

- Part 1: https://ibacity-my.sharepoint.com/:v/g/personal/saliraza_iba_edu_pk/EUi9CXoaskJOq856Wa1n0s8Bghu3Lnsg1sOJv5jAAJTE0A?e=0LG0es
- Part 2: https://ibacity-my.sharepoint.com/:v/g/personal/saliraza_iba_edu_pk/EbK_wAiBhBBHhCRXIQEZe1sBOTjbKslpBX5E-lmSDTLMnw?e=E5sVjq
- Part 3: <video1076155897.mp4>